

## SAN ANTONIO OESTE, Argentina

WMO: 877840

Lat: 40.757S

Lon: 65.031W

Elev: 20

StdP: 101.08

Time Zone: -3.00 (W03)

Period: 95-09

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -3.3       | -1.6 | -11.2                         | 1.4 | 13.6 | -9.0 | 1.7 | 11.7 | 15.3                  | 12.4 | 13.6 | 9.0  | 2.9                   | 320  | 0.660 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 12.2                   | 35.2            | 17.9 | 33.3 | 17.3 | 31.3 | 16.7 | 20.6                | 28.6 | 19.6 | 27.3 | 18.8 | 26.5 | 8.8                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 13.2 | 23.1 | 17.1 | 12.2 | 22.5 | 16.1 | 11.5 | 21.6 | 59.5          | 28.4 | 56.2 | 27.3 | 53.4 | 26.4 | 26.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 13.8          | 12.4 | 11.5 | DB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A   | N/A        | N/A   | N/A  | N/A |
| (5)               |      |      | WB                         | -7.2 | 23.3               | 1.8 | 1.1                                                | -8.5 | 24.1       | -9.5 | 24.7       | -10.5 | 25.3       | -11.8 | 26.1 |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.8   | 22.6  | 21.7  | 19.1  | 14.4  | 10.4  | 7.8   | 7.6   | 9.1   | 11.1  | 15.2  | 18.3  | 20.7  |
|                                                                        | DBStd       | 6.43   | 3.32  | 4.00  | 3.94  | 3.72  | 3.15  | 3.15  | 3.49  | 3.35  | 3.93  | 3.81  | 3.94  | 3.54  |
|                                                                        | HDD10.0     | 299    | 0     | 0     | 0     | 4     | 33    | 78    | 91    | 57    | 32    | 3     | 0     | 0     |
|                                                                        | HDD18.3     | 1752   | 4     | 11    | 38    | 128   | 245   | 315   | 333   | 284   | 219   | 111   | 47    | 17    |
|                                                                        | CDD10.0     | 2057   | 390   | 328   | 283   | 137   | 47    | 13    | 16    | 31    | 66    | 166   | 249   | 331   |
|                                                                        | CDD18.3     | 468    | 136   | 106   | 62    | 11    | 1     | 0     | 0     | 0     | 2     | 15    | 47    | 89    |
|                                                                        | CDH23.3     | 5046   | 1388  | 1126  | 616   | 122   | 8     | 0     | 0     | 3     | 31    | 208   | 578   | 967   |
|                                                                        | CDH26.7     | 2068   | 613   | 526   | 224   | 25    | 1     | 0     | 0     | 0     | 4     | 59    | 221   | 394   |
| (14) Wind                                                              | WSAvg       | 5.7    | 6.6   | 6.4   | 5.9   | 5.4   | 4.9   | 5.1   | 5.5   | 5.4   | 5.2   | 5.8   | 6.2   | 6.4   |
| (15) Precipitation                                                     | PrecAvg     | 260    | 18    | 25    | 34    | 30    | 26    | 19    | 20    | 16    | 15    | 25    | 16    | 18    |
|                                                                        | PrecMax     | 471    | 69    | 123   | 138   | 212   | 114   | 76    | 84    | 81    | 71    | 128   | 67    | 71    |
|                                                                        | PrecMin     | 117    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | PrecStd     | 88     | 17    | 28    | 37    | 38    | 26    | 18    | 20    | 18    | 17    | 26    | 14    | 17    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.1  | 38.1  | 34.8  | 29.5  | 24.2  | 18.8  | 20.5  | 23.4  | 27.3  | 31.5  | 35.4  | 36.8  |
|                                                                        |             | MCWB   | 18.1  | 19.6  | 17.6  | 16.3  | 12.6  | 10.6  | 11.7  | 12.2  | 12.1  | 14.9  | 17.3  | 17.8  |
|                                                                        | 2%          | DB     | 34.7  | 34.9  | 31.6  | 26.6  | 20.2  | 16.3  | 17.5  | 19.7  | 23.4  | 28.0  | 31.6  | 33.7  |
|                                                                        |             | MCWB   | 17.7  | 18.3  | 17.6  | 15.1  | 12.2  | 9.4   | 9.7   | 10.9  | 11.4  | 13.8  | 15.7  | 17.2  |
|                                                                        | 5%          | DB     | 32.6  | 32.6  | 29.1  | 24.3  | 18.1  | 15.0  | 15.8  | 17.7  | 20.8  | 25.5  | 29.1  | 31.1  |
|                                                                        |             | MCWB   | 17.0  | 17.8  | 16.5  | 14.0  | 11.4  | 9.3   | 8.9   | 9.6   | 10.7  | 12.8  | 15.4  | 16.2  |
|                                                                        | 10%         | DB     | 30.1  | 30.0  | 26.9  | 21.6  | 16.3  | 13.5  | 14.2  | 15.9  | 18.7  | 23.0  | 26.6  | 28.9  |
|                                                                        |             | MCWB   | 16.0  | 17.4  | 16.1  | 13.1  | 10.7  | 8.6   | 8.3   | 8.9   | 9.9   | 12.0  | 14.3  | 16.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.9  | 22.5  | 21.1  | 18.7  | 14.9  | 12.4  | 13.6  | 13.5  | 14.2  | 17.6  | 19.6  | 20.9  |
|                                                                        |             | MCDB   | 29.6  | 29.7  | 29.6  | 25.5  | 19.6  | 16.0  | 16.8  | 18.8  | 21.2  | 24.3  | 28.6  | 28.3  |
|                                                                        | 2%          | WB     | 19.6  | 21.0  | 19.5  | 16.7  | 13.5  | 11.2  | 11.4  | 11.9  | 12.7  | 15.4  | 17.8  | 19.7  |
|                                                                        |             | MCDB   | 27.9  | 29.3  | 26.5  | 23.2  | 17.7  | 14.0  | 15.7  | 18.4  | 19.8  | 23.3  | 27.2  | 27.1  |
|                                                                        | 5%          | WB     | 18.7  | 19.9  | 18.4  | 15.1  | 12.6  | 10.2  | 10.0  | 10.6  | 11.7  | 14.2  | 16.5  | 18.5  |
|                                                                        |             | MCDB   | 27.6  | 27.6  | 25.5  | 20.3  | 16.4  | 13.0  | 14.1  | 15.8  | 18.8  | 22.1  | 25.4  | 26.0  |
|                                                                        | 10%         | WB     | 18.0  | 18.9  | 17.4  | 14.1  | 11.7  | 9.3   | 8.7   | 9.7   | 10.7  | 13.2  | 15.5  | 17.5  |
|                                                                        |             | MCDB   | 26.2  | 26.0  | 24.2  | 19.7  | 15.2  | 12.6  | 13.2  | 14.2  | 17.0  | 20.9  | 23.6  | 25.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.2  | 12.1  | 11.6  | 11.8  | 10.5  | 9.7   | 10.8  | 11.3  | 11.7  | 12.0  | 12.5  | 12.3  |
|                                                                        |             | MCDBR  | 17.5  | 17.4  | 16.1  | 15.7  | 13.2  | 11.3  | 12.7  | 13.5  | 15.3  | 16.9  | 17.7  | 18.1  |
|                                                                        |             | MCWBR  | 6.1   | 6.6   | 7.2   | 7.2   | 7.2   | 6.4   | 7.3   | 7.3   | 7.4   | 7.3   | 6.6   | 6.8   |
|                                                                        | 5% WB       | MCDBR  | 12.9  | 13.4  | 12.5  | 12.0  | 10.9  | 9.1   | 10.6  | 11.6  | 13.0  | 13.3  | 13.3  | 12.0  |
|                                                                        |             | MCWBR  | 6.1   | 6.1   | 6.8   | 6.8   | 7.0   | 5.8   | 7.0   | 7.1   | 7.1   | 7.2   | 6.3   | 6.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.395 | 0.380 | 0.363 | 0.351 | 0.331 | 0.312 | 0.316 | 0.333 | 0.363 | 0.363 | 0.369 | 0.392 |
|                                                                        | taud        |        | 2.339 | 2.394 | 2.442 | 2.448 | 2.480 | 2.537 | 2.511 | 2.465 | 2.367 | 2.390 | 2.389 | 2.342 |
|                                                                        | Ebn at Noon |        | 933   | 920   | 891   | 830   | 780   | 771   | 791   | 839   | 875   | 927   | 953   | 942   |
|                                                                        | Edh at Noon |        | 131   | 118   | 103   | 89    | 73    | 63    | 70    | 85    | 109   | 118   | 124   | 132   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 8.05  | 6.85  | 5.27  | 3.62  | 2.21  | 1.76  | 2.01  | 2.84  | 4.27  | 5.84  | 7.35  | 8.17  |
|                                                                        | RadStd      |        | 0.33  | 0.29  | 0.24  | 0.22  | 0.20  | 0.14  | 0.18  | 0.23  | 0.33  | 0.31  | 0.31  | 0.37  |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | +0.28 | N/A     | N/A    | +0.60   | +0.66 | +0.56 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page